



SZABIST
UNIVERSITY
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Final Year Project Long Proposal

Alzheimer Disease Detection using hybrid deep learning

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Revision History

Compiled By	Checked By	Date	Description	Version
Shahmir Hassan	Mr Ahsan Abbas	Date	Revised Section	1.0
Syed Saad Ahmed	Mr Ahsan Abbas	Date	Revised Section	1.1
Ayesha Zaheer	Mr Ahsan Abbas	Date	Revised Section	1.2
Populate	this	for	all	Sections

1. Project Stream

☒ Web-Based FYP ☐ Desktop Application

☐ Mobile App ☐ Game

☐ Hardware-Based

2. Introduction

Alzheimer's disease is a progressive neurological disorder that affects memory, thinking ability, and daily functioning, and it has become one of the major healthcare challenges worldwide. Early diagnosis is critical for slowing disease progression and improving patient quality of life; however, accurate detection at early stages remains difficult. Current diagnostic practices rely heavily on manual interpretation of MRI scans by specialists, which is time-consuming, subjective, and dependent on expert availability. Subtle structural changes in brain images may remain unnoticed, particularly during the early phases of the disease, creating a need for automated and reliable diagnostic support systems.

The objective of this project is to develop a hybrid deep learning-based system for automated Alzheimer's disease detection using brain MRI images. The proposed solution utilizes Convolutional Neural Networks (CNN) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for disease pattern learning, followed by a classification module for stage-wise prediction. The system also incorporates explainable AI visualization using Grad-CAM to highlight important brain regions influencing the model's decision. In addition, a web-based platform is developed that allows users to upload MRI images, view prediction results, interact with a chatbot assistant, and receive AI-based treatment recommendations generated through generative AI techniques.

The proposed system contributes to the growing adoption of artificial intelligence in healthcare by demonstrating an efficient and practical approach to medical image analysis. By supporting early detection and improving diagnostic efficiency, the system also provides features such as prediction history storage, patient monitoring, and user feedback collection through a secure web interface. This project aligns with current trends in AI-driven healthcare solutions and provides a foundation for future research and development in intelligent medical decision-support systems.

3. Project Description

Alzheimer's disease is a progressive neurological disorder that gradually affects memory, reasoning, and daily functioning, making early diagnosis essential for effective treat-

ment and patient care. However, manual analysis of brain MRI scans is time-consuming and highly dependent on expert interpretation, which may lead to delayed or inconsistent diagnosis, especially in early stages where changes are subtle. This project addresses the need for an automated and reliable system that can assist healthcare professionals in detecting Alzheimer’s disease efficiently. The primary objective of the proposed system is to develop a hybrid deep learning–based web application capable of analyzing MRI images and performing stage-wise classification of Alzheimer’s disease. The expected outcomes include accurate prediction results, reduced diagnostic effort, and improved consistency in decision support through automated analysis.

From a technical perspective, the system will be developed using Python and deep learning frameworks such as TensorFlow or PyTorch for model implementation. The proposed architecture combines Convolutional Neural Networks (CNN) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for disease pattern learning, followed by a classification layer for final prediction. Explainable AI techniques such as Grad-CAM will be used to generate visual explanations highlighting important regions influencing model decisions. The backend will be implemented using FastAPI to manage communication between the model and the web interface, while the frontend will be developed using HTML, CSS, and JavaScript for user interaction. MongoDB will be used for storing user information and prediction records. In addition, the system will include features such as a chatbot assistant for user guidance, AI-based treatment recommendation generation, patient monitoring support, and user feedback collection. These components will work together to provide a web-based system where users can upload MRI images, receive automated prediction results, view visual explanations of the prediction, and obtain supportive recommendations through an interactive interface.

4. Real World Problem

Alzheimer’s disease has become one of the most significant neurological disorders affecting the aging population worldwide, creating serious challenges for healthcare systems, patients, and caregivers. Early diagnosis plays a crucial role in slowing disease progression and improving patient quality of life; however, detecting Alzheimer’s disease at an early stage remains difficult. Hospitals and diagnostic centers primarily rely on manual interpretation of MRI brain scans performed by neurologists and radiologists. This process is time-consuming, requires high expertise, and may lead to variability in diagnosis due to human subjectivity. In many regions, especially in developing healthcare environments, limited availability of specialists further delays timely evaluation and treatment.

The problem becomes more critical because early symptoms of Alzheimer’s disease often resemble normal aging, making accurate identification challenging without detailed analysis. Existing automated solutions frequently rely on single deep learning models,

which may fail to capture complex patterns present in MRI images, resulting in reduced accuracy and reliability. These limitations affect multiple stakeholders, including patients who may receive delayed diagnosis, healthcare professionals facing increasing workloads, and healthcare institutions managing growing patient numbers.

If this issue remains unresolved, late detection may continue to increase treatment costs, reduce the effectiveness of medical interventions, and place additional emotional and financial burdens on families. Therefore, there is a strong need for an intelligent and automated system that can assist professionals by analyzing MRI images efficiently and providing consistent, stage-wise predictions for Alzheimer's disease detection. Such a system should also support visual interpretation of model predictions using explainable AI techniques and provide accessible web-based tools that allow users to upload MRI scans, view results, and obtain supportive guidance through an interactive platform.

5. Problem Statement

Alzheimer's disease is a progressive neurological disorder that significantly affects memory, cognitive ability, and daily functioning, particularly among the aging population. The increasing number of Alzheimer's patients has created a major challenge for healthcare systems, as early diagnosis remains difficult and heavily dependent on expert interpretation of brain MRI scans. Manual analysis is time-consuming, subject to human variability, and may fail to detect subtle structural changes during the early stages of the disease. Delayed or inaccurate diagnosis reduces the effectiveness of treatment and increases the emotional, social, and economic burden on patients, families, and healthcare providers.

The primary problem addressed in this project is the lack of an efficient and reliable automated system capable of accurately detecting Alzheimer's disease from MRI images at an early stage. Existing approaches often rely on single-model deep learning solutions that may not effectively capture complex patterns present in brain images, leading to limited accuracy and inconsistent results. Therefore, there is a need for a hybrid deep learning-based solution that can improve detection accuracy, support stage-wise classification, and assist healthcare professionals by providing consistent and timely decision support. This project aims to address this problem by developing an intelligent web-based system that integrates CNN and LSTM models for MRI analysis, explainable AI techniques such as Grad-CAM for visual interpretation, and an interactive platform for automated diagnosis and user support.

6. Names of existing similar FYPs (if repeated)

N/A

7. New Functionalities (main) to be added Features

N/A

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8. Application Review

8.1 AIRAMed

- **Key Features:**

- Automated Brain Volume Analysis: Uses MRI scans to measure structural brain changes associated with Alzheimer’s disease.
- AI-Based Image Processing: Applies artificial intelligence techniques for detecting abnormal brain patterns.
- Clinical Research Support: Designed primarily for neurological and medical research.
- MRI Data Analysis: Focuses on identifying disease-related structural variations.

- **Limitations:**

- Limited Explainability: Does not clearly explain prediction reasoning.
- No Stage-wise Classification: Limited support for disease progression analysis.
- Complex Usage: Requires technical expertise for operation.

8.2 NeuGRID

- **Key Features:**

- Neuroimaging Data Processing: Supports large-scale MRI data analysis for research purposes.
- Research Infrastructure: Provides computational support for neurological studies.
- Collaborative Research Platform: Enables sharing and processing of neuroimaging datasets.
- Advanced Data Handling: Designed for handling large medical datasets.

- **Limitations:**

- Complex Setup: Difficult for non-technical users to operate.

- Research-Oriented Design: Not intended for direct clinical prediction.
- Limited User Interaction: Lacks simple and user-friendly interface.

8.3 ADNI Pipeline

- **Key Features:**

- Standard Alzheimer Dataset Usage: Widely used for academic Alzheimer’s research.
- MRI-Based Analysis: Supports structural brain imaging analysis.
- Research Benchmarking: Used for evaluating different research models.
- Data-Driven Evaluation: Enables comparison of experimental results.

- **Limitations:**

- No Integrated Application Interface: Not designed for end users.
- Limited Explainable AI Support: Does not provide visual reasoning.
- Research Focused: Not suitable for real-time prediction systems.

8.4 Neuro Vista (Proposed)

- **Key Features:**

- Hybrid CNN + LSTM Architecture: Combines spatial feature extraction and disease progression modeling.
- Multi-Stage Classification: Performs stage-wise Alzheimer’s disease prediction.
- Explainable AI (Grad-CAM): Provides heatmap-based visual explanations highlighting important brain regions.
- Web-Based Interface: Allows easy MRI upload and automated prediction.
- Prediction Confidence Score: Provides probability-based output for decision support.
- User-Friendly Design: Suitable for academic and healthcare usage.

Table 1 presents a comparative analysis of key features among various applications. The comparison includes functionalities crucial for effective management. The comparison aims to highlight the unique offerings of each application and underscore the comprehensive capabilities of the Proposed System.

Table 1: Applications Comparison

Features	Applications			
	AIRAMed [1]	NeuGRID [2]	ADNI Pipeline [3]	Proposed System
Hybrid CNN + LSTM	✗	✗	✗	✓
Multi-Stage Classification	✗	✗	✗	✓
Attention Mechanism / Heatmap	✗	✗	✗	✓
XAI with reasoning	✗	✗	✗	✓
Web Based Interface	✗	✓	✗	✓
Deployment Ready	✓	✗	✗	✓
Affordable for Academic Use	✗	✓	✓	✓
Prediction Confidence	✗	✗	✗	✓

9. Scope

The scope of this project defines the boundaries, objectives, and expected deliverables of the proposed Alzheimer’s disease detection system. The project focuses on developing a hybrid deep learning–based application capable of analyzing brain MRI images for automated Alzheimer’s detection and stage-wise classification. The system is designed as a decision-support tool to assist healthcare professionals by providing reliable prediction results through a web-based interface. The scope includes dataset preparation, model development, system implementation, and performance evaluation. However, the project is limited to research and decision-support purposes and does not aim to replace professional medical diagnosis or clinical treatment processes.

9.1 Included Aspects

- Automated analysis of brain MRI images using a hybrid deep learning approach (CNN + LSTM)
- Alzheimer detection and stage-wise classification based on dataset categories
- Development of a web-based interface for MRI upload and prediction result visualization
- Explainable AI visualization using Grad-CAM to highlight important brain regions
- AI-based recommendation generation and chatbot interaction for user support
- Model training, performance evaluation, and system testing

9.2 Excluded Aspects

- Real-time hospital or clinical system integration
- Direct medical diagnosis or prescription of treatment
- Use of multimodal medical data such as PET scans or clinical history
- MRI image acquisition or hardware-level medical imaging processes

9.3 Modules Description

Module 1: Dataset Preprocessing and Data Augmentation

- Data collection
- Image preprocessing
- Data augmentation

Module 2: Frontend Development

- Image upload
- User authentication

Module 3: Database Integration

- User data storage
- Prediction history

Module 4: Disease Detection Using Hybrid Deep Learning

- CNN-based feature extraction from MRI images
- LSTM-based pattern learning
- Alzheimer disease prediction

Module 5: User Management and Authentication

- User registration
- User authentication

Module 6: Chatbot System

- User interaction
- Chatbot integration

Module 7: Backend Development and Deployment

- API development
- Backend services

- System scalability

Module 8: Disease Explanation Using Explainable AI

- Grad-CAM heatmap generation
- Visualization
- Real-time Explanation

Module 9: Treatment Recommendation Using Generative AI

- Symptom analysis
- Recommendation Engine
- Real-time updates

Module 10: Patient Monitoring and Follow-Up

- System monitoring
- Patient follow-up

Module 11: Model Testing and Deployment

- Model testing
- Model deployment

Module 12: User Feedback

- Feedback collection
- System improvement

10. Modules

Table 2 provides a breakdown of project modules and assigned students based on deadlines, showcasing their roles and responsibilities.

Table 2: Modules Breakdown

Deadline	Modules	Student
P1 Mid	Dataset Preprocessing and Data Augmentation	Shahmir Hassan (22108204)
	Frontend Development	Syed Saad Ahmed (22108210)
	Database Integration	Ayesha Zaheer (23108278)

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Table 2 – Continued from previous page

Deadline	Modules	Student
P1 Final	Disease Detection Using Deep Learning	Shahmir Hassan (22108204)
	User Management and Authentication	Syed Saad Ahmed (22108210)
	Chatbot System	Ayesha Zaheer (23108278)
P2 Mid	Backend Development and Deployment	Shahmir Hassan (22108204)
	Disease Explanation Using Explainable AI	Syed Saad Ahmed (22108210)
	Treatment Recommendation Using Generative AI	Ayesha Zaheer (23108278)
P2 Final	Patient Monitoring and Follow-Up	Shahmir Hassan (22108204)
	Model Testing and Deployment	Syed Saad Ahmed (22108210)
	User Feedback	Ayesha Zaheer (23108278)

11. Development Environment

Tools: Python, TensorFlow, PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, Grad-CAM, FastAPI, HTML, CSS, JavaScript, Visual Studio Code, Jupyter Notebook, Git, GitHub

DBMS: My SQL

12. Significance

Alzheimer’s disease represents a major global healthcare challenge due to its increasing prevalence and the absence of a definitive cure. As aging populations continue to grow worldwide, the number of individuals affected by dementia is increasing rapidly, placing significant pressure on healthcare systems, caregivers, and families. Early detection of Alzheimer’s disease is critical because timely intervention can help slow disease progression, improve patient quality of life, and support better treatment planning. However, traditional diagnosis based on manual interpretation of MRI scans is time-consuming and highly dependent on expert availability, which can lead to delays and inconsistencies in diagnosis. This highlights the importance of intelligent automated systems that can assist healthcare professionals in early and reliable disease detection.

The proposed project contributes to the healthcare and technological domains by

introducing a hybrid deep learning–based system for automated Alzheimer’s detection and stage-wise classification. By combining CNN-based feature extraction with LSTM-based disease progression modeling, the system improves prediction accuracy and provides structured analysis of MRI images. The integration of Explainable Artificial Intelligence through Grad-CAM further enhances transparency by visually explaining model decisions, thereby increasing trust and usability for medical and research applications. The system also improves efficiency by reducing manual analysis time, minimizing human error, and providing faster prediction results through a web-based interface. In addition, the platform supports user interaction through chatbot assistance and AI-based recommendation generation to guide users after prediction results.

In the long term, the project supports the growing trend of AI-assisted healthcare and digital medical transformation. The proposed system aligns with modern advancements in intelligent diagnostics, medical image analysis, and data-driven decision-making. Its scalable design allows future expansion toward integration with larger medical datasets, enhancement through improved models, and extension to other neurological disorders. By promoting early detection and efficient resource utilization, the project contributes to sustainable healthcare practices and provides a foundation for future research and development in AI-based medical diagnosis systems.

13. Tools and Techniques

The proposed Alzheimer’s disease detection system utilizes modern artificial intelligence and web development technologies to ensure accurate analysis, efficient processing, and user-friendly interaction. The system is designed to combine deep learning–based medical image analysis with a lightweight web-based interface, enabling automated MRI analysis and prediction generation. Image processing and model development techniques are applied to extract meaningful features from MRI scans, while explainable AI methods improve transparency by providing visual reasoning for model predictions. These technologies collectively support the development of a reliable and accessible decision-support system for Alzheimer’s disease detection.

- **Front-End:** The user interface is developed using HTML, CSS, and JavaScript to provide an interactive and user-friendly environment. The interface allows users to upload MRI images, view prediction results, and visualize model outputs in a clear and structured format.
- **Back-end:** Server-side logic is implemented using FastAPI in Python to handle communication between the user interface and the deep learning model. The backend processes user requests, performs model inference, and efficiently returns classification results through API communication.
- **Database:** The system utilizes MySQL for efficient data storage and management.

The database stores user information, prediction history, and MRI image records, enabling secure data handling and easy retrieval for analysis purposes.

- **Tools:** The project utilizes Python as the primary programming language along with TensorFlow and PyTorch for deep learning model development. OpenCV is used for image preprocessing, while NumPy and Pandas support data handling and processing. Scikit-learn is used for model evaluation and performance analysis. Grad-CAM is applied for explainable AI visualization. Development and experimentation are carried out using Visual Studio Code and Jupyter Notebook, while Git and GitHub are used for version control and collaborative development.

References

- [1] H. Almalki, A. O. Khadidos, N. Alhebaishi, and E. M. Senan, “Early detection of Alzheimer’s disease progression stages using hybrid of CNN and transformer encoder models,” *Scientific Reports*, vol. 15, Art. no. 16799, May 2025, doi: 10.1038/s41598-025-01072-5.
- [2] S. Zolfaghari, A. Joudaki, and Y. Sarbaz, “A hybrid learning approach for MRI-based detection of Alzheimer’s disease stages using dual CNNs and ensemble classifier,” *Scientific Reports*, vol. 15, Art. no. 25342, Jul. 2025, doi: 10.1038/s41598-025-11743-y.
- [3] Q. Wu, Y. Wang, X. Zhang, H. Zhang, and K. Che, “A hybrid transformer-based approach for early detection of Alzheimer’s disease using MRI images,” *BioImpacts*, vol. 15, 2025.
- [4] S. A. and A. D., “An explainable web-based diagnostic system for Alzheimer’s disease using XRAI and deep learning on brain MRI,” *Diagnostics*, vol. 15, no. 20, 2025.